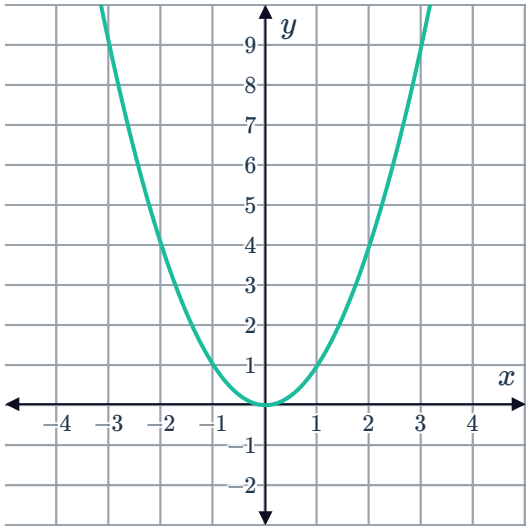
Graphs of parabolas

1

a

x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9

b



c No

d $x = 0$

e 0

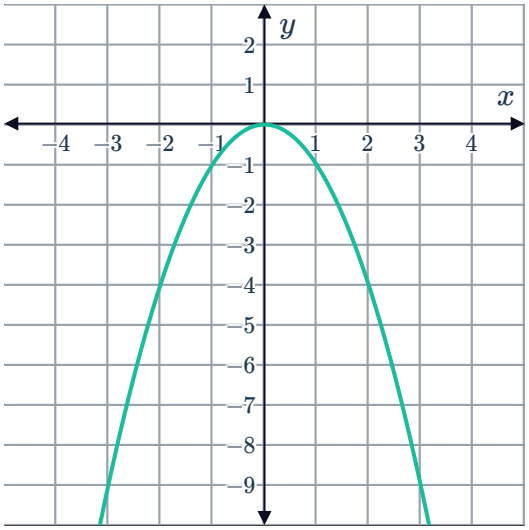
f 2

2

a

x	-3	-2	-1	0	1	2	3
y	-9	-4	-1	0	-1	-4	-9

b



c No

d $x = 0$

e 0

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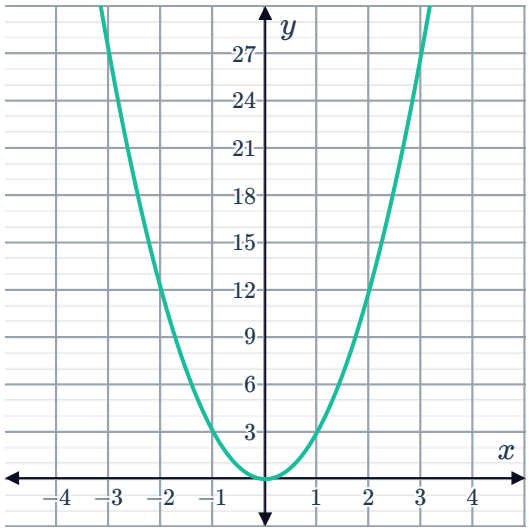
3

a

i

x	-3	-2	-1	0	1	2	3
y	27	12	3	0	3	12	27

ii

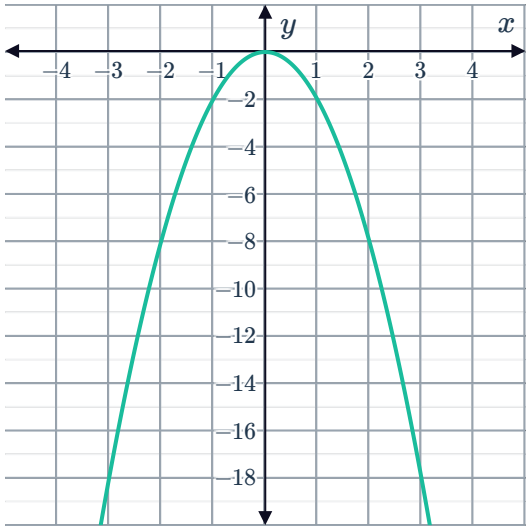


b

i

x	-3	-2	-1	0	1	2	3
y	-18	-8	-2	0	-2	-8	-18

ii



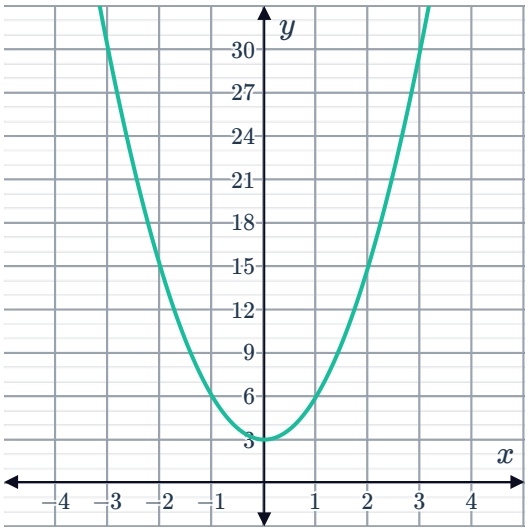
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c

i

x	-3	-2	-1	0	1	2	3
y	30	15	6	3	6	15	30

ii

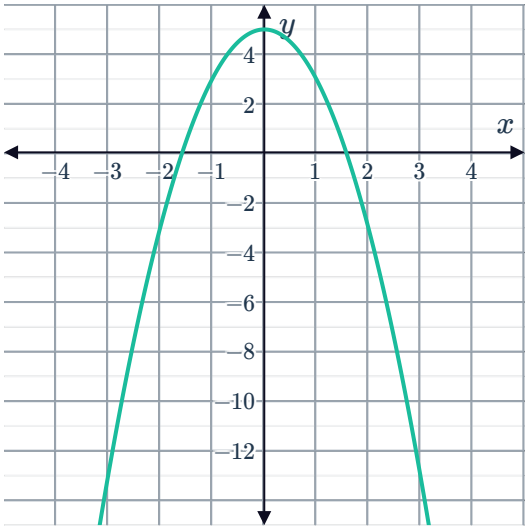


d

i

x	-3	-2	-1	0	1	2	3
y	-13	-3	3	5	3	-3	-13

ii



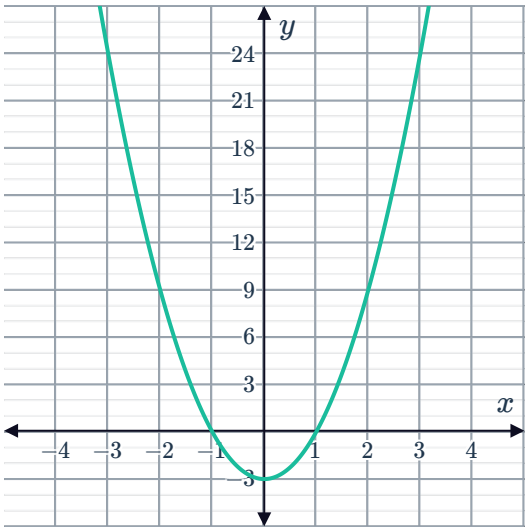
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e

i

x	-3	-2	-1	0	1	2	3
y	24	9	0	-3	0	9	24

ii

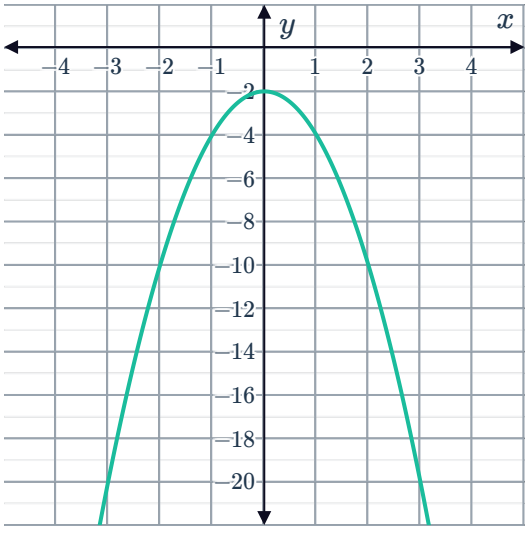


f

i

x	-3	-2	-1	0	1	2	3
y	-20	-10	-4	-2	-4	-10	-20

ii



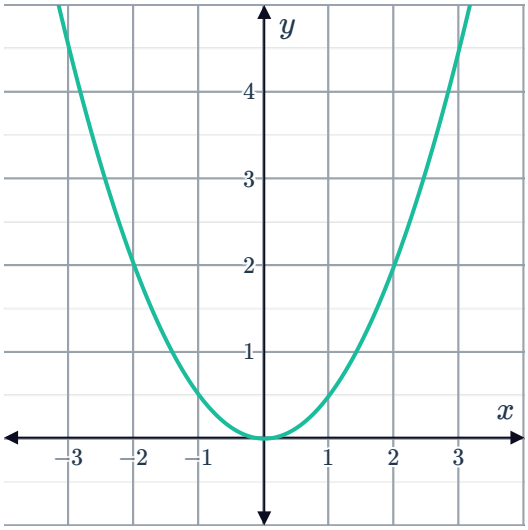
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g

i

x	-3	-2	-1	0	1	2	3
y	$\frac{9}{2}$	2	$\frac{1}{2}$	0	$\frac{1}{2}$	2	$\frac{9}{2}$

ii

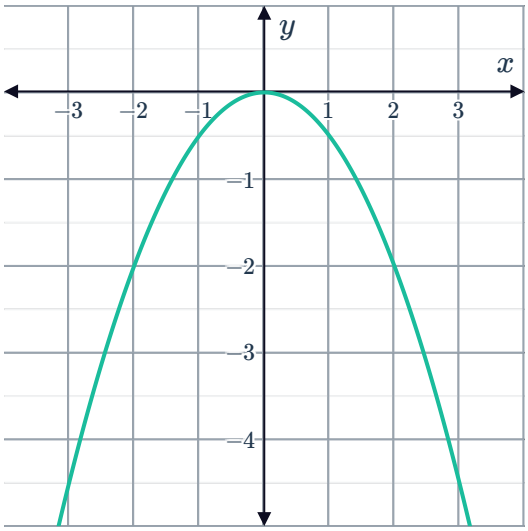


h

i

x	-3	-2	-1	0	1	2	3
y	$-\frac{9}{2}$	-2	$-\frac{1}{2}$	0	$-\frac{1}{2}$	-2	$-\frac{9}{2}$

ii



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Transformations of parabolas

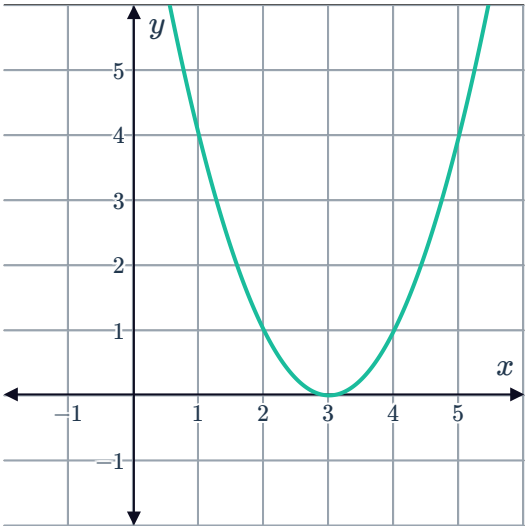
4

a

i

x	1	2	3	4	5
y	4	1	0	1	4

ii



iii 0

iv $x = 3$

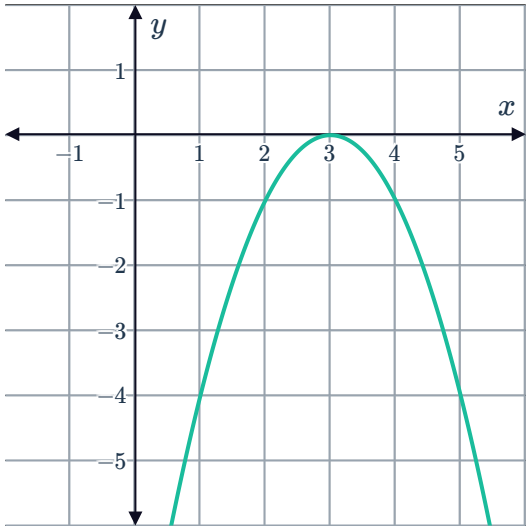
v $(3, 0)$

b

i

x	1	2	3	4	5
y	-4	-1	0	-1	-4

ii



iii 0

iv 3

v $(3, 0)$

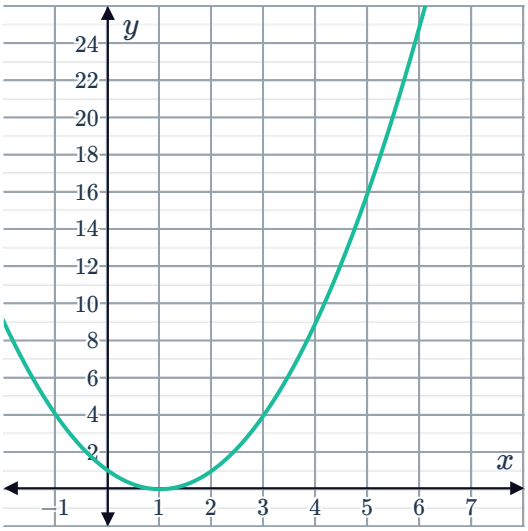
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c

i

x	1	2	3	4	5
y	0	1	4	9	16

ii



iii 0

iv 1

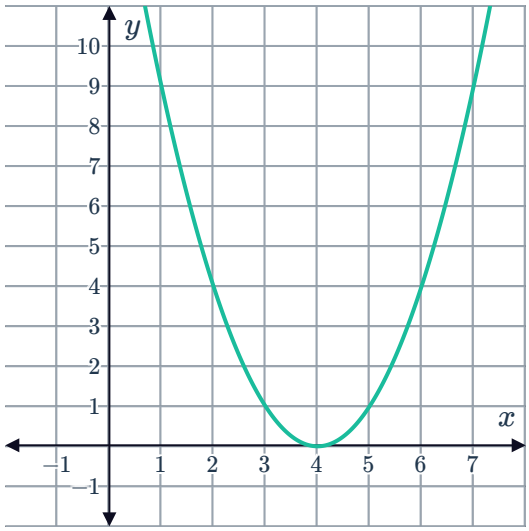
v $(1, 0)$

d

i

x	1	2	3	4	5
y	9	4	1	0	1

ii



iii 0

iv 4

v $(4, 0)$

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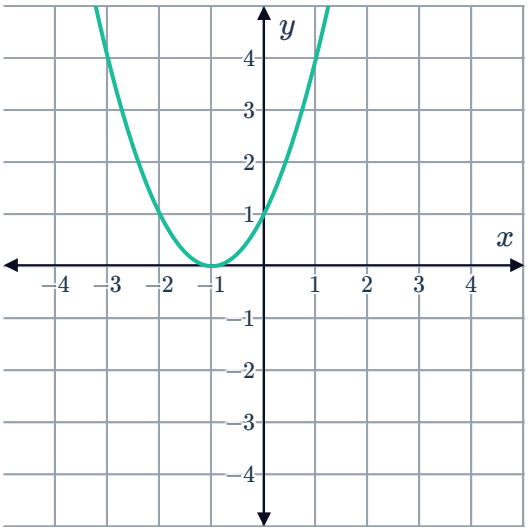
5

a

i

x	-3	-2	-1	0	1
y	4	1	0	1	4

ii



iii 0

iv $x = -1$

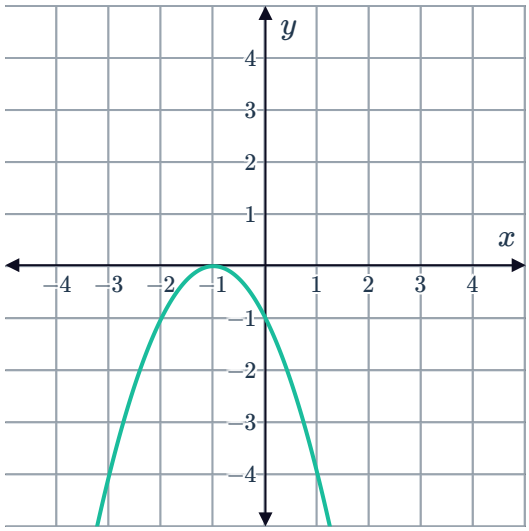
v $(-1, 0)$

b

i

x	-3	-2	-1	0	1
y	-4	-1	0	-1	-4

ii



iii 0

iv -1

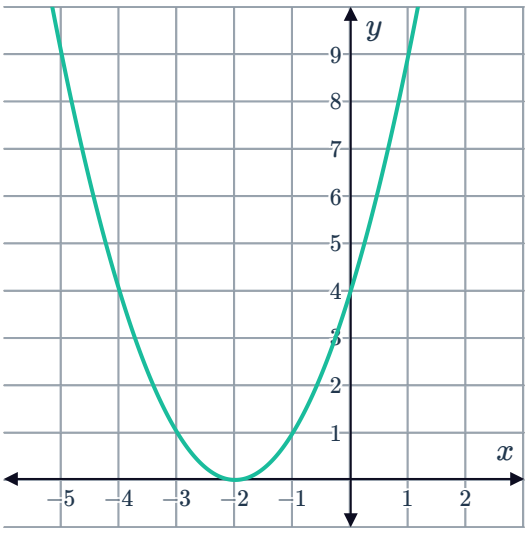
v $(-1, 0)$

c

i

x	-3	-2	-1	0	1
y	1	0	1	4	9

ii



iii 0

iv -2

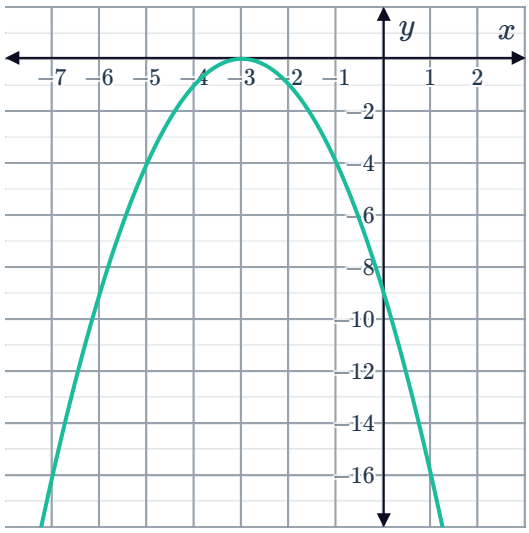
v $(-2, 0)$

d

i

x	-3	-2	-1	0	1
y	0	-1	-4	-9	-16

ii



iii 0

iv -3

v $(-3, 0)$

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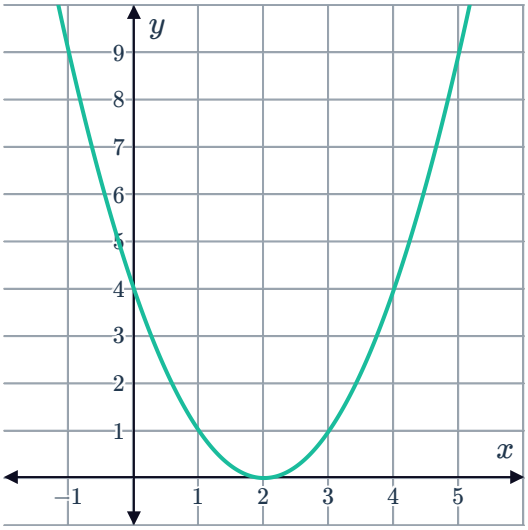
6

a

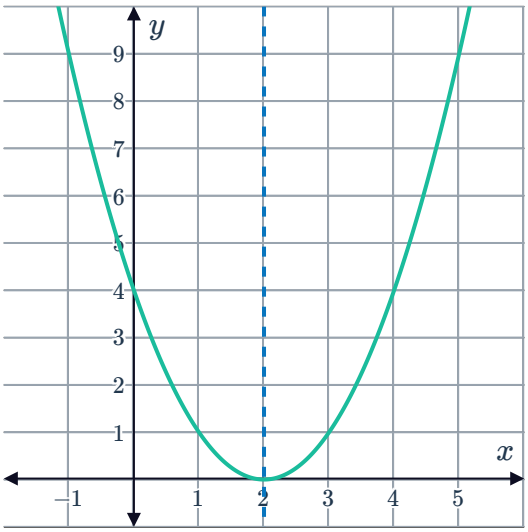
i $(2, 0)$

ii $x = 2$

iii



iv

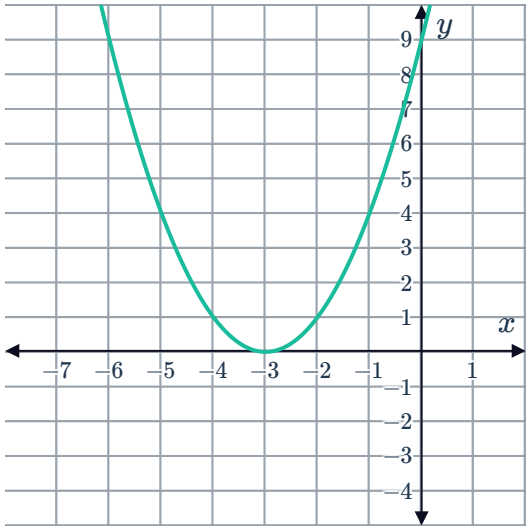


b

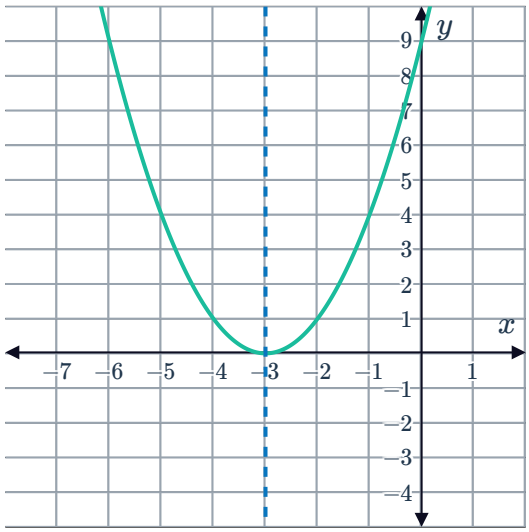
i $(-3, 0)$

ii $x = -3$

iii



iv



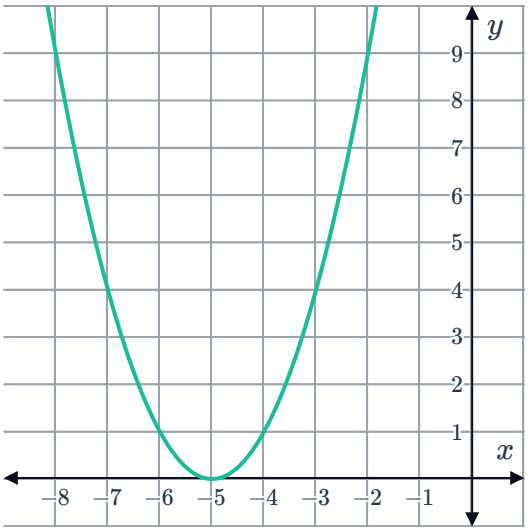
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c

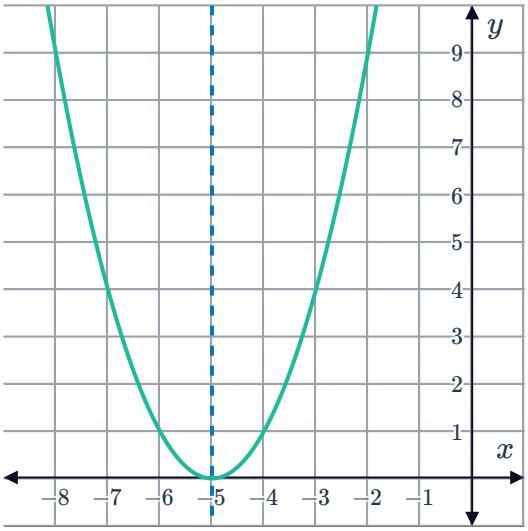
i $(-5, 0)$

ii $x = -5$

iii



iv

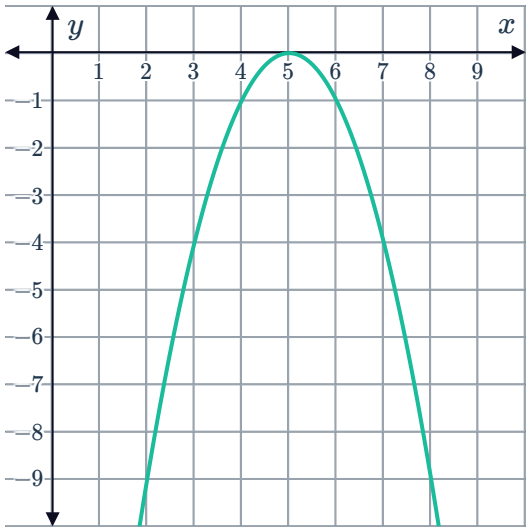


d

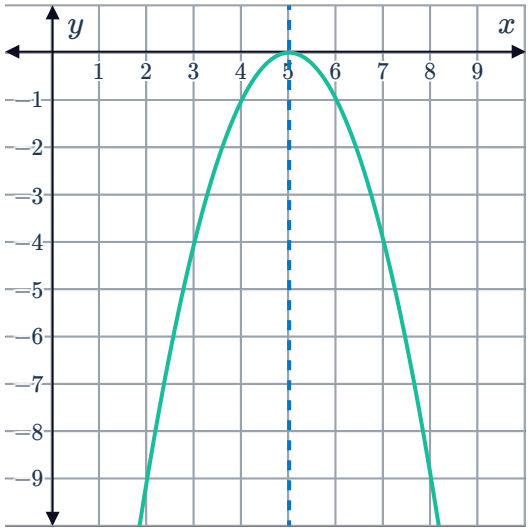
i $(5, 0)$

ii $x = 5$

iii



iv



7 Horizontally translated by -4 units.

8

a Concave up **b** Concave down **c** Concave down **d** Concave up

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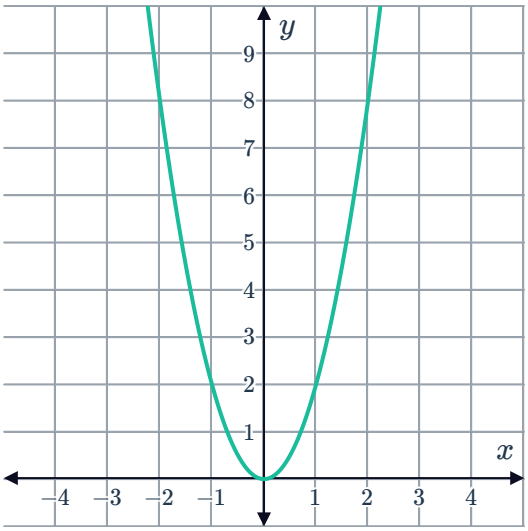
9

a

i

x	-2	-1	0	1	2
y	8	2	0	2	8

ii



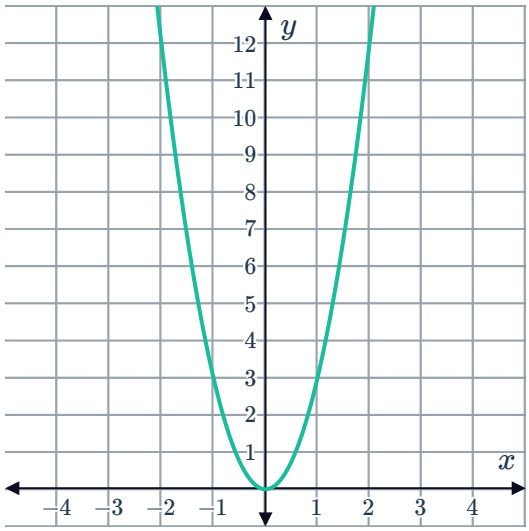
iii $(0,0)$

b

i

x	-2	-1	0	1	2
y	12	3	0	3	12

ii



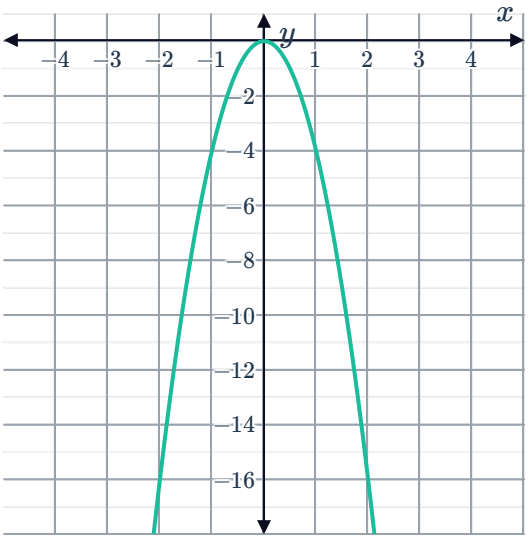
iii $(0,0)$

c

i

x	-2	-1	0	1	2
y	-16	-4	0	-4	-16

ii



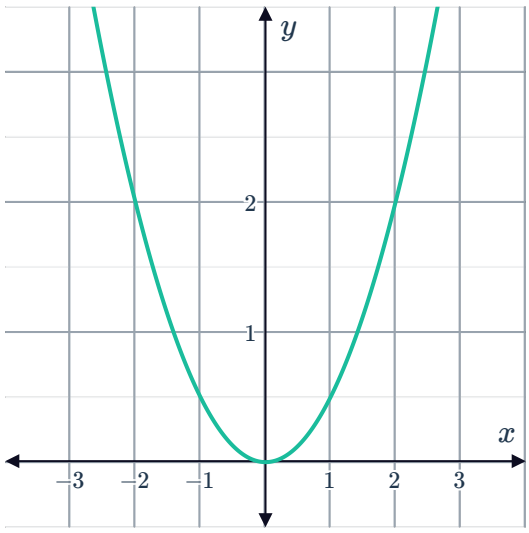
iii $(0,0)$

d

i

x	-2	-1	0	1	2
y	2	$\frac{1}{2}$	0	$\frac{1}{2}$	2

ii



iii $(0,0)$

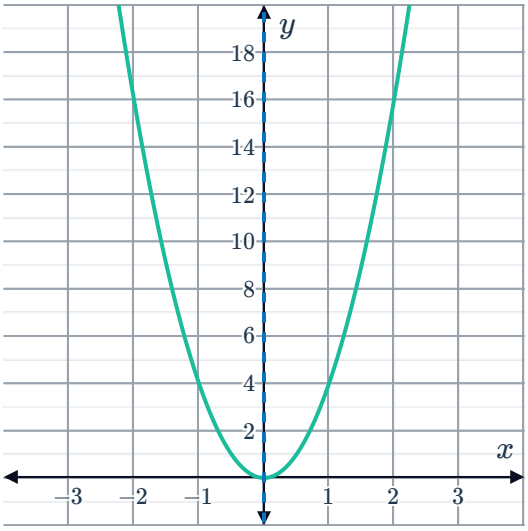
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10

a

i $x = 0$

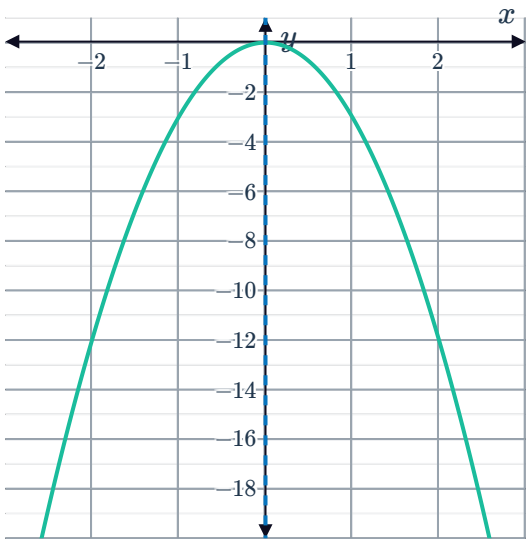
ii



b

i $x = 0$

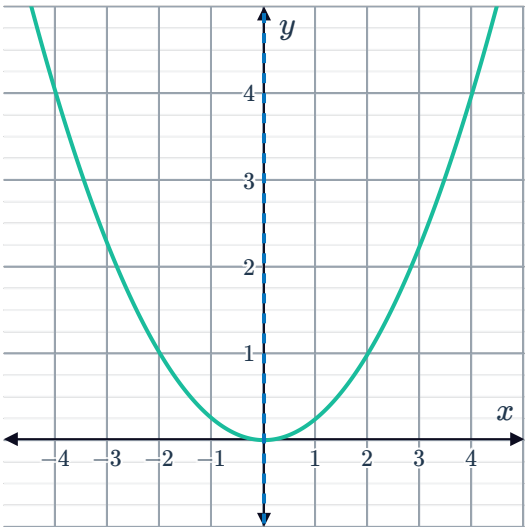
ii



c

i $x = 0$

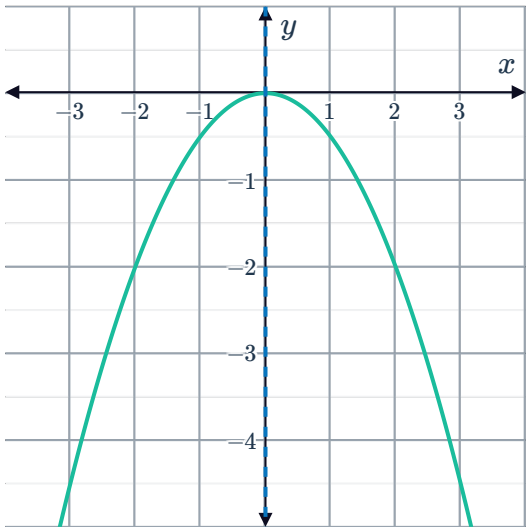
ii



d

i $x = 0$

ii



11 Vertically scaled by 5 units.

12

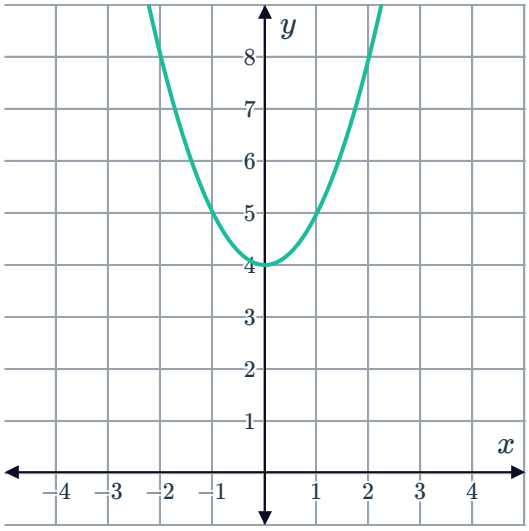
a $y = x^2 + 3$ **b** $y = x^2 - 2$ **c** $y = (x - 5)^2$ **d** $y = (x + 4)^2$

e $y = 2x^2$ **f** $y = -3x^2$ **g** $y = \frac{1}{2}x^2$ **h** $y = -x^2$

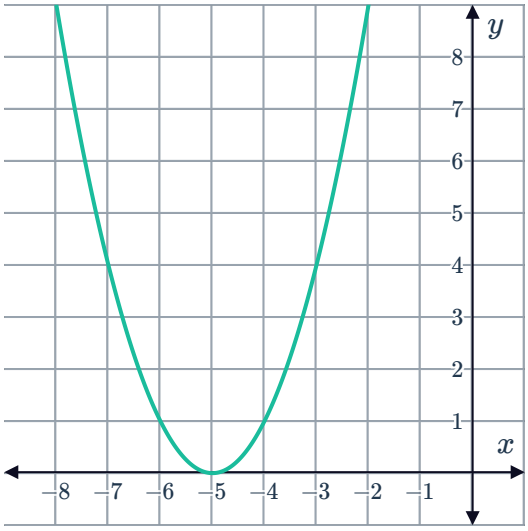
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13

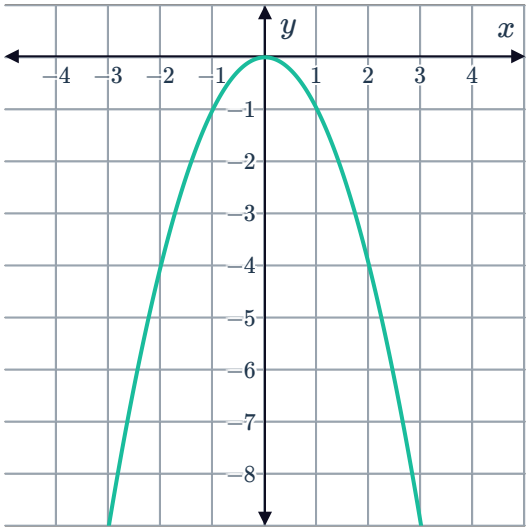
a



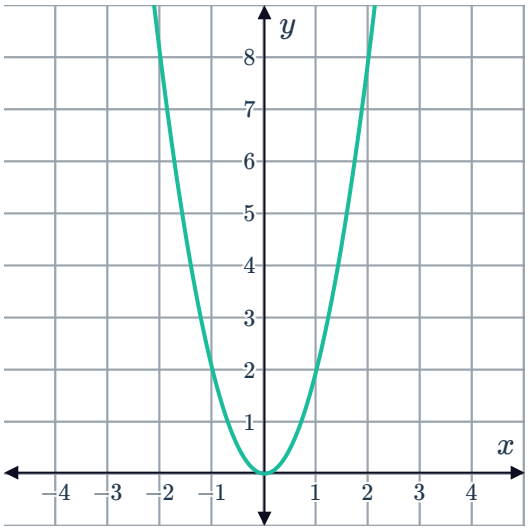
b



c



d



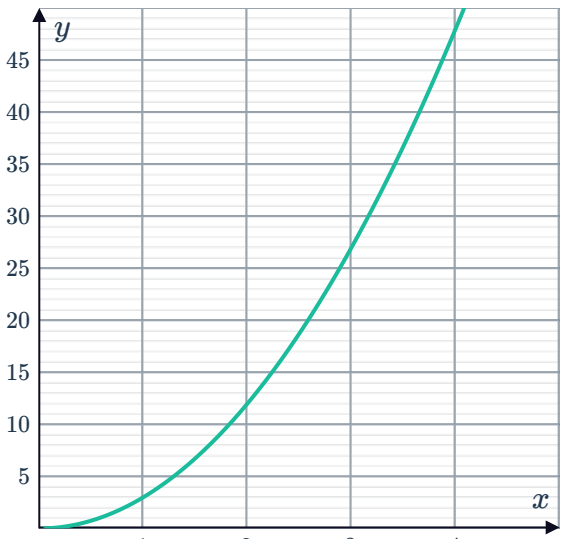
Applications

14

a

x (m)	0	1	2	3	4
y (m)	0	3	12	27	48

b



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c 0 m

d $x = 0$

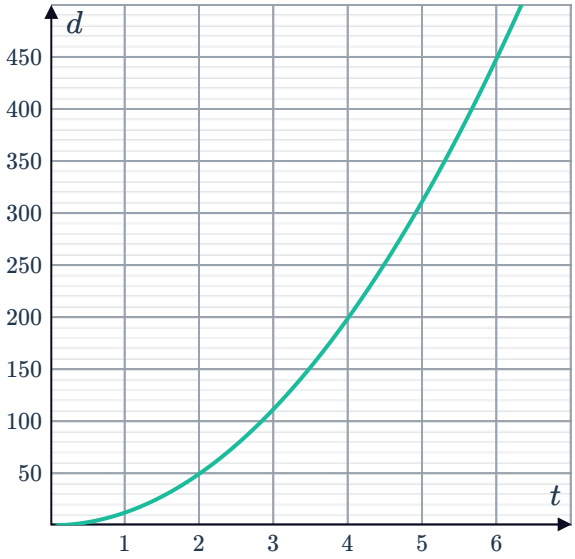
e $(0, 0)$

15

a

time (t)	0	2	4	6
distance (d)	0	50	200	450

b



c $t = 3\text{ s}$

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